

Model Curriculum

IoT – Solution Architect

SECTOR: IT-ITeS
SUB-SECTOR: FUTURE SKILLS
OCCUPATION: INTERNET OF THINGS
REF ID: SSC/Q8202, V1.0
NSQF LEVEL: 7

 Skill India कौशल भारत - कुशल भारत	 IT - ITes SSC NASSCOM	 N-S-D-C National Skill Development Corporation Transforming the skill landscape
Certificate		
COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS		
is hereby issued by the IT-ITes Sector Skills Council NASSCOM		
for MODEL CURRICULUM		
Complying to the National occupation standards of Job Role / Qualification Pack: 'IoT – Solution Architect' QP No. 'SSC/Q8202 NSQF Level 7'		
Date of Issuance: March 11th 2019 Valid Upto *: March 11th 2020 * Valid up to the next review date of the Qualitification Pack		 Authorised Signatory (IT-ITes Sector Skills Council NASSCOM)

TABLE OF CONTENTS

1. Curriculum	01
2. Trainer Prerequisites	17
3. Assessment Criteria	18

IoT - Solution Architect

CURRICULUM / SYLLABUS

This program is aimed at training candidates for the job of a “IoT - Solution Architect”, in the “IT-ITes” Sector/Industry and aims at building the following key competencies amongst the learner

Program Name	IoT - Solution Architect		
Qualification Pack Name and Reference ID.	SSC/Q8202, V1.0		
Version No.	1.0	Version Update Date	11/03/2019
Pre-requisites to Training	Bachelor's Degree in Engineering / Technology / Statistics / Mathematics / Computer Science		
Training Outcomes	After completing this programme, participants will be able to: • Explain the nature of work across the IT-ITes sector, the various sub sectors and their evolution. • Elaborate the various occupations under the Future Skills sub sector and the impact of these on organizations and businesses. • Discuss the evolution of IoT and evaluate the possible impact of IoT on businesses and society • List common security and privacy risks that affect IoT solutions and methods that mitigate them • Apply principles and methods of enterprise architecture which conform to industry standards and frameworks • Implement data management standards within the organization and the globe • Compare different platforms on which solution is to be deployed, based on solution requirements and architecture constraints. • Assess the implications of business and technical requirements on solution architecture design and develop solution architecture using appropriate design standards, methods and tools. • Maintain solution architecture by reviewing technical capabilities roadmap, measuring and optimizing performance and monitoring changes in solution design standards • Design security architectures across IoT solutions • Perform selection of components to be integrated into the IoT solution • Develop visualizations by evaluating various tools and dashboards and applying storytelling and design principles • Design UI/UX visualizations by developing scenarios, flows and user journeys to represent them		

	• Perform Value Engineering and Analysis of IoT solution components • Perform integrations across IT and OT systems • Test and validate architecture of IoT solutions • Evaluate and upgrade architecture of IoT solutions • Evaluate standards and ensure compliance of IoT solutions • Integrate IoT solution across its various layers while ensuring its reliability, scalability and availability • Create various types of technical documents • Build relationships with stakeholders by establishing rapport, listening actively and providing continuous updates
--	--



This course encompasses 13 out of 13 National Occupational Standards (NOS) of “IoT - Solution Architect” Qualification Pack issued by “IT-ITeS Sector Skills Council”.

Sr. No.	Module	Key Learning Outcomes	Equipment Required
1	IT-ITeS/BPM Industry – An Introduction Theory Duration (hh:mm) 06:00 Practical Duration (hh:mm) 03:00 Corresponding NOS Code Bridge Module	- Explain the relevance of the IT-ITeS sector - State the various sub-sectors in the IT-ITeS sector - Detail the nature of work performed across the sub-sectors - List organizations in the sector - Discuss the evolution of the sub sectors and the way forward - Explain the disruptions happening across the IT-ITeS sector	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
2	Future Skills – An Introduction Theory Duration (hh:mm) 01:00 Practical Duration (hh:mm) 01:00 Corresponding NOS Code Bridge Module	- Discuss the general overview of the Future Skills sub-sector - Describe the profile of the Future Skills sub-sector - Explain the various occupations under this sub-sector - List key trends across the occupations in this sub-sector - List various roles in the Future Skills sub-sector	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
3	Internet of Things - An Introduction Theory Duration (hh:mm) 06:00 Practical Duration (hh:mm) 03:00 Corresponding NOS Code Bridge Module	- Define the term "Internet of Things" - Discuss the evolution of IoT and the trends that have led to it - Evaluate the possible impact of IoT on businesses and society - Analyze existing IoT use cases and applications across industries - List prospective IoT use cases and applications within your area of expertise	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
4	IoT Security and Privacy Theory Duration (hh:mm)	List common security and privacy risks that affect IoT solutions	- Whiteboard and Markers - LCD Projector and Laptop for presentations

	06:00 Practical Duration (hh:mm) 03:00 Corresponding NOS Code Bridge Module	- Analyze instances of security risks to real-world IoT use cases - List technologies and methods that mitigate security risks to IoT solutions - List privacy standards and regulations that mitigate security risks to IoT solutions - List social and privacy impacts caused by proliferation of IoT solutions	- Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
5	Basics of Architecture Theory Duration (hh:mm) 08:00 Practical Duration (hh:mm) 12:00 Corresponding NOS Code Bridge Module	- Theorize enterprise architecture principles and illustrate their translation to practice - Construct architecture which conform to industry standards and frameworks - Identify critical success factors for common enterprise architect approaches - Recognize and use principles and methods applicable to enterprise architecture - Evaluate alternative systems architectures on the basis of organizational strategy and enterprise requirements	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
6	Global Regulations and Standards Theory Duration (hh:mm) 13:00 Practical Duration (hh:mm) 17:00 Corresponding NOS Code Bridge Module	- Identify general principles and basic concepts of data management standards across the globe - Identify the key actors under the regulations and understand their roles - Evaluate the rights of data owners - Evaluate various enforcement and compliance mechanisms - Act in accordance with enforcement and compliance obligations	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Chart paper and sketch pens - Internet with Wi-Fi (Min 2 Mbps Dedicated)
7	Solution Architecture – Platforms Theory Duration (hh:mm)	- Evaluate different platforms on which the solution is to be deployed - Assess solution architecture constraints, i.e., deploying a	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following:



	25:00 Practical Duration (hh:mm) 50:00 Corresponding NOS Code SSC/N8113	solution on web vs mobile platforms - Assess the performance of the application on different platforms - Perform impact analyses to review the uses and limitations of deploying the solution across different platforms - Select an appropriate platform for the deployment of the solution - Determine the solution architecture based on the type of platforms such as a Thin Clients (for online applications), Thick Clients (for offline applications) or Smart Clients (for occasionally connected applications) - Stay updated on the vendor's solution capabilities and product roadmaps - Develop competencies to guide developers in resolving their most challenging issues	- PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
8	Solution Architecture – Design Theory Duration (hh:mm) 25:00 Practical Duration (hh:mm) 51:00 Corresponding NOS Code SSC/N8113	- Define system level architecture for the solution - Develop solution architectures using appropriate design standards, methods and tools - Translate component specifications into detailed designs for implementation - Build architectures that are distributed and highly scalable - Monitor performance of the solution architecture - Assess the implications of the business, technical and data requirements on the design of solution architecture	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
9	Solution Architecture – Maintenance Theory Duration (hh:mm) 25:00 Practical Duration	- Implement different approaches to measure and optimize performance of solution architectures - Review technical capabilities roadmap for existing solution architectures	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)

	(hh:mm) 50:00 Corresponding NOS Code SSC/N8114	• Migrate and update solution architectures based on the need • Monitor changes in solution design standards • Monitor metrics on an on-going basis to ensure the architecture is scalable in the long term • Develop best practices to ensure any changes to the architecture is not impacting the existing customer	
10	Security Architecture Theory Duration (hh:mm) 25:00 Practical Duration (hh:mm) 50:00 Corresponding NOS Code SSC/N8210	• Discuss the basic principles of security design and architecture • Discuss ways to capture security requirements for the desired solution • Comprehend practices employed to assess the state of IoT devices and components • Identify security design frameworks to be implemented across the IoT stack • Evaluate suitability of security design frameworks to be implemented in the IoT solution • Identify different types of security threats and risks associated with IoT solutions • Discuss security standards and protocols that are essential to creating a security architecture • Discuss standard security safeguards and controls that should be incorporated to secure networks • Demonstrate ways to prevent unauthorized access to IoT systems and sub-systems • Identify appropriate tests that can be leveraged to ensure IoT devices and components perform as per expectations • Demonstrate best practices to implement IoT security architecture • Demonstrate ways to identify, align and implement security monitoring models that ensure	• Whiteboard and Markers • LCD Projector and Laptop for presentations • Lab equipped with the following: • PCs/Laptops • Internet with Wi-Fi (Min 2 Mbps Dedicated)



		security anomalies are identified at the appropriate instance • Discuss practices for upgrading and maintaining security architecture as and when found necessary	
11	Technology Selection Theory Duration (hh:mm) 15:00 Practical Duration (hh:mm) 30:00 Corresponding NOS Code SSC/N8211	• Identify the best practices to determine technology trends in IoT • Discuss the market dynamics surrounding the proliferation of IoT solutions • Discuss the impact of IoT on the design and implementation of technology solutions • Comprehend current literature on emerging IoT technologies, practices and standards • Perform technology selection of solutions and components to integrate into the IoT solution • Identify criteria to evaluate suppliers of IoT solutions and components • Demonstrate how to track suppliers and their solutions • Demonstrate how to define processes and metrics to continuously evaluate the quality of supplied solutions and components • Discuss how to identify and build strategic alliances with suitable suppliers • Demonstrate how to select hardware and software components to be integrated with IoT solution • Define methodologies to assess if IoT solutions meet the desired requirements	• Whiteboard and Markers • LCD projector and laptop for presentations • Lab equipped with the following: • PCs/Laptops • Internet with Wi-Fi (Min 2 Mbps Dedicated)
12	Developing Visualizations Theory Duration (hh:mm) 10:00	• Discuss techniques that will help identify the visualization needs of the target audience • Evaluate the various visualization tools and dashboards and	• Whiteboard and Markers • LCD projector and laptop for presentations • Lab equipped with the following:

	Practical Duration (hh:mm) 15:00 Corresponding NOS Code SSC/N8212	distinguish between their pros and cons - Identify appropriate visualization tools and dashboards based on needs of the target audience - Discuss the various stages of the data lifecycle including acquisition, backup, maintenance, usage and destruction of data - Discuss how to use colour and imagery to making interfaces function clearly and seamlessly - Discuss how to apply design principles, human perception and effective storytelling while developing visualizations	- PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
13	UI/UX Design Theory Duration (hh:mm) 15:00 Practical Duration (hh:mm) 35:00 Corresponding NOS Code SSC/N8212	- Discuss the stages of the UI/UX development process, from user research to defining project scope and developing sitemaps and wireframes - Discuss the challenges of research, planning, setting goals, understanding the user, structuring content, and developing interactive sequences - Discuss the atomic design methodology and its five distinct stages - Develop scenarios, flows, alerts and notifications and user journeys to represent visualizations - Design UI/UX for different devices and interfaces such as websites and mobile apps - Develop UI/UX designs for Application Programming Interfaces - Discuss how to plan and conduct usability sessions - Develop wireframes, visual mockups and clickable prototypes	- Whiteboard and Markers - LCD projector and laptop for presentations - Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
14	Value Engineering and Analysis	Define the fundamental concepts, principles and phases	Whiteboard and Markers

	Theory Duration (hh:mm) 25:00 Practical Duration (hh:mm) 50:00 Corresponding NOS Codes SSC/N8213	of Value Engineering and Analysis - Evaluate frameworks for applying Value Engineering in IoT solutions - Evaluate regulations that must be complied with while implementing Value Engineering frameworks - Evaluate standards that must be complied with while implementing Value Engineering frameworks - Demonstrate how to gather information about needs and constraints of IoT solution - Perform cost-function analysis to improve solution value while maintaining or reducing costs - Develop cost-function matrices to identify functions with high cost-function ratio - Discuss the Quality Function Deployment (QFD) methodology and develop value analysis matrices - Develop Function Analysis System Technique (FAST) diagrams to identify critical project components - Perform project risk and scenario analysis - Evaluate interoperability of solutions, i.e., the ability for the solution to work together regardless of their varying technical specifications - Discuss how to reuse components as much as possible while performing Value Engineering and Analysis - Document the results of Value Engineering and Analysis processes for their implementation	- LCD projector and laptop for presentations - Lab equipped with the following: - PCs/Laptops Internet with Wi-Fi (Min 2 Mbps Dedicated)
15	Information Technology – Operational Technology (IT-OT) Integrations	Discuss the definition, history and future potential of Information Technology – Operational Technology (IT-OT) integrations	- Whiteboard and Markers - LCD projector and laptop for presentations

	Theory Duration (hh:mm) 25:00 Practical Duration (hh:mm) 50:00 Corresponding NOS Code SSC/N8214	- Discuss business scenarios where IT-OT integrations can be implemented - Develop Risk Management Plans to evaluate the viability of IT-OT integrations - Perform cost-benefit analysis to evaluate the viability of IT-OT integrations - Perform vendor selection, if necessary, for implementation of IT-OT integrations - Develop data models required for interchange of data between IT and OT systems - Perform integrations of middleware between IT and OT systems - Discuss how to extract data from disparate systems to facilitate analytics - Establish common and consistent semantics for the systems to be integrated - Develop relevant documentation to implement IT-OT integration - Develop the frequency and the process for review and upgradation of integrated IT-OT systems	- Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
16	Architecture Testing Theory Duration (hh:mm) 15:00 Practical Duration (hh:mm) 30:00 Corresponding NOS Code SSC/N8215	- Discuss the basic concepts of solution architecture testing - Identify the different approaches (such as unit testing, integration testing, systems testing, acceptance testing etc.) to test solution architecture - Perform usability tests on solution architecture to test ease of use of architecture - Perform compatibility tests on solution architecture to test the compatibility of different software, middleware, storage systems and network systems - Perform regression tests on the architecture to ensure that changes made to the solution	- Whiteboard and Markers - LCD projector and laptop for presentations - Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)



		architecture do not hinder the functioning of other parts of the solution • Perform scalability tests to ensure architecture can handle the projected increase in user traffic and data volume • Perform reliability tests to ensure architecture can perform failure-free operations for a specified period of time in a specified environment • Perform simulation tests to model its operations of the architecture	
17	Architecture Evaluation Theory Duration (hh:mm) 15:00 Practical Duration (hh:mm) 30:00 Corresponding NOS Code SSC/N8216	• Review the architecture on an ongoing basis to identify opportunities for optimization • Discuss the differences between evaluating architecture early or late in its development • Identify the different categories of architecture evaluation • Identify the different methods to evaluate architecture • Evaluate the performance of the architecture • Evaluate the reliability of the architecture • Describe the different security standards and protocols essential for a secure solution architecture • Evaluate the security levels of the architecture • Evaluate the portability of solution architecture • Evaluate the modifiability of solution architecture • Perform regular upgrades to IoT architecture, as and when required	• Whiteboard and Markers • LCD projector and laptop for presentations • Lab equipped with the following: • PCs/Laptops • Internet with Wi-Fi (Min 2 Mbps Dedicated)
18	Architecture Standards and Compliance Theory Duration (hh:mm)	• Discuss the different types of IoT architecture frameworks including both open-source and proprietary frameworks	• Whiteboard and Markers • LCD projector and laptop for presentations • Lab equipped with the following:



	15:00 Practical Duration (hh:mm) 30:00 Corresponding NOS Code SSC/N8217	- Identify common architectural standards used across both the organization and the industry - Identify the different criteria for evaluating architectural standards - List the different types of security risks associated with solution architectures - Assess if architectural standards meet the security requirements - Discuss the various industrial, technological and organizational regulations to be complied with while creating a secure architecture design - Identify different risk management frameworks to identify risks associated with the solution architecture - Apply risk management frameworks to analyze risks to the IoT solution - Create Risk Management Plans that can mitigate potential risks to architectures	- PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)
19	IoT Solution Integration Theory Duration (hh:mm) 25:00 Practical Duration (hh:mm) 50:00 Corresponding NOS Code SSC/N8220	- Discuss the significance of rapid integration capabilities in IoT solutions - Perform technology selection of software/hardware solutions to integrate into the IoT solution - Demonstrate how to perform integration across various solution components such as devices, networks, databases and cloud infrastructure - Develop strategies to collect, store and analyze data from various integrated devices - Develop metrics to compare perform solution performance both before and after integrations - Evaluate reliability and availability of architecture both before and after integrations - Demonstrate how to build scalable integrations, i.e.,	- Whiteboard and Markers - LCD projector and laptop for presentations - Lab equipped with the following: - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated)



		integrations which allow the evolution of the architecture throughout its lifecycle • Demonstrate how to build flexible integrations, i.e., integrations which allow the support of multiple devices, protocols and data models • Demonstrate how to build architectures that enable third party integration services including databases, cloud platforms and analytics services • Evaluate interoperability of the integrated solution, i.e., the ability for components of the integrated solution to work together regardless of their varying technical specifications • Evaluate various standards and ensure that the integrated solution complies with the identified standards	
20	Technical Documentation Theory Duration (hh:mm) 10:00 Practical Duration (hh:mm) 15:00 Corresponding NOS Code SSC/N8238	• Comprehend the significance of technical documentation • Identify different sources for obtaining information for creating technical documentation • Identify different standard templates to create a documentation plan • Identify ways to define purpose and scope of the documentation • Evaluate best practices on how to plan, write, and deliver technical documentation • Identify practices for writing technical documents so that they are clear and concise • Demonstrate how to create various types of technical documents such as online configuration manuals, user guides, installation guides, etc. • Identify best practices to review, maintain and update technical documents	• Whiteboard and Markers • LCD Projector and Laptop for presentations • Lab equipped with the following: • PCs/Laptops • Internet with Wi-Fi (Min 2 Mbps Dedicated)



		Identify best practices to be followed before, during and after publishing technical documents	
21	Stakeholder Management Theory Duration (hh:mm) 10:00 Practical Duration (hh:mm) 15:00 Corresponding NOS Code SSC/N9012	- Define the needs and perspectives of the stakeholders in order to build consensus - Employ active listening behaviors while communicating with stakeholders - Build rapport and collaborate with the stakeholders - Manage the expectations of the stakeholders, including quality and performance expectations - Provide continuous updates on project/activity status and changes in timelines - Evaluate the fundamentals of negotiations such as negotiating positions, BATNA (Best Alternative to a Negotiated Agreement) and integrative and distributive negotiations - Identify causes of conflict and methods to resolve conflict	- Whiteboard and Markers - LCD Projector and Laptop for presentations - Provision to write emails and send in the lab - Lab with provision for internet, email, word processor and presentation software - Chart paper, markers, picture magazines and old newspapers
	Total Duration: Theory Duration 320:00 Practical Duration 590:00	Unique Equipment Required - Whiteboard and Markers - LCD Projector and Laptop for presentations - Chart paper, markers, picture magazines and old newspapers - Lab equipped with the following: - - PCs/Laptops - Internet with Wi-Fi (Min 2 Mbps Dedicated) - Provision for internet, email, word processor and presentation software - Chart paper and sketch pens Popular Software Tools (At least one of the tools listed across each of the five tool categories is required) <u>Development Environment Software:</u> Apache Maven, PowerShell, Docker, Visual Basic, COBOL, Ruby, C <u>Development Software:</u> C++, Python, R, C#, Java, Perl, ABAP, MATLAB	

		<u>Web Development Software:</u> AJAX, LAMP Stack, Nodejs, ReactJS, AngularJS, Elm <u>Mobile Application Development Platform:</u> Apple Xcode, Android Studio, Kony, Appzillon, Telerik, Xamarin <u>IoT Platforms:</u> Watson IoT, GE Predix, PTC Thingworx, AWS IoT, Azure IoT, Eclipse IoT, DeviceHub, Arduino, Raspberry Pi
--	--	--

Grand Total Course Duration: 910 Hours, 0 Minutes

(This syllabus/ curriculum has been approved by SSC: IT- ITes Sector Skills Council NASSCOM)

Trainer Prerequisites for Job role: “IoT - Solution Architect” mapped to Qualification Pack: “SSC/Q8202, V1.0”

Sr. No.	Area	Details
1	Description	To deliver accredited training service, mapping to the curriculum detailed above, in accordance with the Qualification Pack <u>SSC/Q8202, V1.0</u>
2	Personal Attributes	This job may require the individual to work independently and take decisions for his/her own area of work. The individual should have a high level of analytical thinking ability, passion for Artificial Intelligence and Big Data Analytics, and attention for detail, should be ethical, compliance and result oriented, should also be able to demonstrate interpersonal skills, along with willingness to undertake desk-based job with long working hours.
3	Minimum Educational Qualifications	Graduate in any discipline preferably Science/Computer Science/Electronics and Engineering /Information Technology
4a	Domain Certification	Certified for Job Role: “IoT - Solution Architect” mapped to QP: “ <u>SSC/Q8202, V1.0</u> ”. Minimum accepted score is 80%
4b	Platform Certification	Recommended that the trainer is certified for the Job role “Trainer” mapped to the Qualification Pack “ <u>MEP/Q0102</u> ”. Minimum accepted score is 80% aggregate
5	Experience	5+ years of work experience/internship in DevOps Engineer or related roles

Criteria For Assessment Of Trainees

Job Role IoT - Solution Architect

Qualification Pack SSC/Q8202, V1.0

Sector Skill Council IT-ITeS

Guidelines for Assessment

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
6. To pass a QP, a trainee should score an average of 70% across generic NOS' and a minimum of 70% for each technical NOS
7. In case of *unsuccessful completion*, the trainee may seek reassessment on the Qualification Pack.



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
1. SSC/N8113 Design new solution architectures as per specifications	PC1. identify business requirements	100	5	2	3
	PC2. evaluate the platform on which the solution is to be built		7	2	5
	PC3. evaluate changes in cross-border data governance issues, PII (Personally Identifiable Information) regulations, GDPR (General Data Protection Regulations), global-scale, security, and monitoring		5	2	3
	PC4. select appropriate design standards, methods and tools		7	2	5
	PC5. define system level architecture for the solution		8	3	5
	PC6. design data feeds, data quality checks and components needed to configure the solution		10	3	7
	PC7. define detailed component specifications		5	2	3
	PC8. translate component specifications into detailed designs for implementation		5	1	4
	PC9. undertake impact analysis on major design options		8	3	5
	PC10. develop technical capabilities roadmap for new solution architectures		8	3	5
	PC11. create documentation on designed solution		5	0	5



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	architectures for appropriate people				
	PC12. validate solution architectures with appropriate people		5	0	5
	PC13. identify data quality metrics and guide processes that track and measure data quality continually		8	3	5
	PC14. troubleshoot and resolve issues with new solution architectures		7	2	5
	PC15. identify potential risks, and define mitigation plans and processes for appropriate people		7	2	5
	Total		100	30	70
2. SSC/N8114 Maintain existing architectures for solutions	PC1. review existing system architecture designs to ensure balance of functional, service quality, and systems management requirements	100	30	10	20
	PC2. review existing system architecture designs to ensure selection of appropriate technology, efficient use of resources, and integration of multiple systems		30	10	20
	PC3. review technical capabilities roadmap for existing solution architectures		20	5	15



Compulsory NOS				Marks Allocation	
Total Marks: 1300					
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC4. monitor changes in solution design standards and update architecture		15	5	10
	PC5. update documentation on existing solution architectures for appropriate people		5	0	5
	Total		100	30	70
3. SSC/N8210 Design and implement security architecture across end-to-end IoT solutions	PC1. acquire deep understanding of all technology systems of the organization	100	25	7.5	17.5
	PC2. identify business and security requirements of the solution		10	3	7
	PC3. identify and adopt security standards and protocols appropriate for the solution		5	1.5	3.5
	PC4. plan, research and design robust security architectures for the solution		5	1.5	3.5
	PC5. implement security architecture across all layers of the solution (such as the device, network and cloud layers)		5	1.5	3.5
	PC6. adopt security safeguards such as routers, firewalls and Identity Access Management tools		10	3	7
	PC7. protect the IoT network from unauthorized access or malicious intent		5	1.5	3.5
	PC8. ensure only authorized devices should		5	1.5	3.5



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	be able to connect to the network				
	PC9. test security architecture to ensure expected behavior as per requirements		5	1.5	3.5
	PC10. design and implement security policies and standards		5	1.5	3.5
	PC11. ensure that solution architecture implements, operates and controls prescribed security policies and standards		5	1.5	3.5
	PC12. ensure that security policies and standards are followed at all times		5	1.5	3.5
	PC13. ensure that solution architecture performs day-to-day security processes such as threat and vulnerability management		5	1.5	3.5
	PC14. continuously update and upgrade security systems as needed		5	1.5	3.5
	Total		100	30	70
4. SSC/N8211 Evaluate new technologies, solutions and vendor products for integration into IoT solutions	PC1. gather and analyze intelligence about products, customers, competitors and landscapes	100	25	7.5	17.5
	PC2. identify emerging IoT technologies, practices and standards		25	7.5	17.5



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC3. stay up to date with latest industry trends		10	3	7
	PC4. track both leading and emerging suppliers across the spectrum of solution providers		15	4.5	10.5
	PC5. evaluate suppliers based on criteria such as breadth of offering, solution performance, geographic availability, support services and security		15	4.5	10.5
	PC6. perform technology selection of software/hardware solutions to integrate into the IoT solution		5	1.5	3.5
	PC7. assess solution fit based on requirements of IoT architecture		5	1.5	3.5
	Total		100	30	70
5. SSC/N8212 Manage visualizations, UI/UX design and data lifecycles for all stages of data lifecycle	PC1. evaluate available data visualization tools to be used for IoT solution	100	25	7.5	17.5
	PC2. select visualization tools depending on company's goals, customer base and type of data being collected		15	4.5	10.5
	PC3. manage dashboards that visualize data coming from multiple devices		15	4.5	10.5
	PC4. develop designs that enhance user experience		10	3	7
	PC5. develop designs which include the ability to		5	1.5	3.5

Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	easily upgrade processing units				
	PC6. design alerts, notifications and their associated actions while developing designs		5	1.5	3.5
	PC7. develop APIs which safely allow a user to connect, control, and read information from devices		10	3	7
	PC8. perform processes of all stages of data lifecycle including acquisition, backup, maintenance, usage and destruction		15	4.5	10.5
	Total		100	30	70
6. SSC/N8213 Perform value engineering and analysis of IoT solution components	PC1. identify components/solutions to be analyzed	100	5	1.5	3.5
	PC2. gather information about components/solutions to be analyzed from relevant stakeholders		5	1.5	3.5
	PC3. perform cost-function analysis of components/solutions		10	3	7
	PC4. identify improvements that either improve functionality or lower cost of components/solutions		10	3	7
	PC5. prepare reports that summarize the conclusions and specific proposals		10	3	7



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC6. describe action plans for implementation of recommendations		10	3	7
	PC7. evaluate possible standards that can be used while designing and optimizing IoT solutions		10	3	7
	PC8. identify and implement appropriate standards across IoT solutions		10	3	7
	PC9. ensure value engineering leads to components/solutions which are compliant with industrial, technological and organizational regulations		10	3	7
	PC10. ensure value engineering leads to interoperability between different components of the IoT solution		10	3	7
	PC11. evaluate existing IoT solution and enhance reusability of IoT solutions and components		10	3	7
	Total		100	30	70
7. SSC/N8214 Integrate Information Technology (IT) systems with Operational Technology (OT) systems	PC1. identify defined areas where IT-OT integration can deliver tangible value	100	10	3	7
	PC2. identify and engage with stakeholders who will be involved in IT-OT integration		10	3	7
	PC3. define scope of the IT-OT integration initiative		5	1.5	3.5

Compulsory NOS				Marks Allocation	
Total Marks: 1300					
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC4. assess costs and benefits of performing IT-OT integration		5	1.5	3.5
	PC5. evaluate risks, consequences and pitfalls of performing IT-OT integration		10	3	7
	PC6. document value of IT-OT integration including potential growth factors such as enabling new business models		5	1.5	3.5
	PC7. assess the need to consult external vendors to perform IT-OT integration		5	1.5	3.5
	PC8. obtain executive buy-in to perform IT-OT integration		10	3	7
	PC9. evaluate and document current state across people, processes and technology systems prior to IT-OT integration		5	1.5	3.5
	PC10. evaluate and document future state across people, processes and technology systems post IT-OT integration		5	1.5	3.5
	PC11. perform IT-OT integration considering factors such as analysis of standards, data ownership and security policies		10	3	7
	PC12. ensure that IT-OT integration does not interfere with daily or mission-critical tasks on both IT and OT systems		5	1.5	3.5



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC13. perform quality testing for IT-OT integrations		5	1.5	3.5
	PC14. determine frequency to review and update integrated systems		5	1.5	3.5
	PC15. review and update integrated systems over regular intervals		5	1.5	3.5
	Total		100	30	70
8. SSC/N8215 Test and validate architecture of IoT solutions	PC1. test usability of IoT architecture with real users	100	15	4.5	10.5
	PC2. test user's ease to use the application, flexibility in handling controls and system's ability to meet its objectives		15	4.5	10.5
	PC3. test performance issues in the design of IoT architecture		15	4.5	10.5
	PC4. test multiple OS versions, browser types, generations of devices, communications nodes, etc. on IoT architecture		15	4.5	10.5
	PC5. perform regression testing to overcome upgrade related issues		15	4.5	10.5
	PC6. use virtualization tools and technologies to perform reliability and scalability testing		10	3	7
	PC7. build IoT test environments which		15	4.5	10.5



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	involve simulation of edge devices				
	Total		100	30	70
9. SSC/N8216 Continuously evaluate and upgrade architecture of IoT solutions	PC1. assess IoT architecture to highlight their characteristics, strengths and weaknesses	100	15	4.5	10.5
	PC2. evaluate performance based on responsiveness of the architecture		10	3	7
	PC3. evaluate reliability of the architecture based on its ability to keep operating over time		10	3	7
	PC4. evaluate availability of the architecture based on the proportion of time the system is up and running		10	3	7
	PC5. evaluate security of the architecture based on its ability to resist unauthorized attempts at usage		15	4.5	10.5
	PC6. evaluate modifiability of architecture speed and costs needed to make changes to it		10	3	7
	PC7. evaluate portability of architecture based on its ability to run under different computing environments		10	3	7



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC8. perform regular upgrades to IoT architecture		10	3	7
	PC9. ensure upgrades address increased scalability, reliability and security demands		10	3	7
	Total		100	30	70
10. SSC/N8217 Use appropriate standards and ensure compliance of IoT solutions	PC1. evaluate what value a given standard brings to IoT architecture	100	20	6	14
	PC2. evaluate risks of not using a particular standard for the architecture		15	4.5	10.5
	PC3. identify standards that best fit the needs of the architecture		15	4.5	10.5
	PC4. ensure solutions are compliant with industry, technological and organizational regulations		15	4.5	10.5
	PC5. understand risk and compliance requirements that must be followed and document them		20	6	14
	PC6. identify plan of action to be proactively manage risks and offset costs of compliance		15	4.5	10.5
	Total		100	30	70
11. SSC/N8220 Integrate IoT solutions across various device components, networks, databases,	PC1. identify the short-term and long-term integration needs of the solution	100	10	3	7
	PC2. implement integration of services such as data integration,		10	3	7

Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
analytics solutions and cloud infrastructure	cloud service integration, business intelligence, etc. depending on needs of the solution				
	PC3. create and coordinate roles, responsibilities, and delivery of integration needs across various teams (both internal and external)		15	4.5	10.5
	PC4. perform integration across various solution components such as devices, networks, databases and cloud infrastructure		20	6	14
	PC5. manage architecture design, security architecture design and hardware design across all components of the solution during integration		10	3	7
	PC6. ensure reliability, scalability, availability of integrated architecture		10	3	7
	PC7. perform IoT standardization to streamline overall solution integration of IoT projects		10	3	7
	PC8. ensure that standardization enables interoperability across devices		10	3	7
	PC9. ensure that standardization leads to reduction in overall project costs		5	1.5	3.5
	Total		100	30	70



Compulsory NOS Total Marks: 1300				Marks Allocation	
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
12. SSC/N8238 Create technical documents and manuals	PC1. identify the purpose and the scope of the activity for which technical documentation is to be produced	100	20	6	14
	PC2. obtain information for the technical document from relevant sources and stakeholders		15	4.5	10.5
	PC3. draft technical document ensuring that content is concise, complete and easy to consume		15	4.5	10.5
	PC4. review technical document content with relevant stakeholders and document owners		10	3	7
	PC5. ensure that technical document is formatted and designed as per specifications		10	3	7
	PC6. transfer technical document to relevant stakeholders for sign-off and publishing		10	3	7
	PC7. continuously review and update technical document		20	6	14
	Total		100	30	70
13. SSC/N9012 Manage and collaborate with stakeholders for project success	PC1. identify the larger business and organizational context behind the requirements of the stakeholder	100	10	3	7
	PC2. manage fluctuating stakeholder priorities and expectations		5	1.5	3.5

Compulsory NOS				Marks Allocation	
Total Marks: 1300					
Assessment outcomes	Assessment Criteria for outcomes	Total Marks	Out Of	Theory	Skills Practical
	PC3. consult stakeholders early in critical organization-wide decisions		10	3	7
	PC4. use formal communication methods to collaborate with stakeholders (such as meetings, conference calls, emails etc.)		5	1.5	3.5
	PC5. keep stakeholders updated on changes in project requirements		10	3	7
	PC6. define the frequency of communication with all the stakeholders		10	3	7
	PC7. use suitable tools to represent numbers and pictures to present details		10	3	7
	PC8. respond to requests in a timely and accurate manner		10	3	7
	PC9. take feedbacks from stakeholders regularly		5	1.5	3.5
	PC10. continuously improve work deliverables/service based on stakeholder feedback		15	5	10
	PC11. plan deliverables based on stakeholder needs		10	3	7
	Total		100	30	70